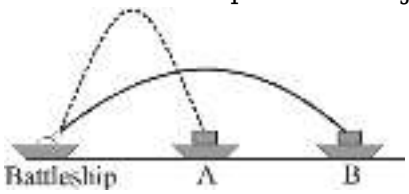
PHYSICS

1) Two guns on a battleship simultaneously fires two shells with same speed at enemy ships. If the shells follow the parabolic trajectories as shown, which ship will get hit first?



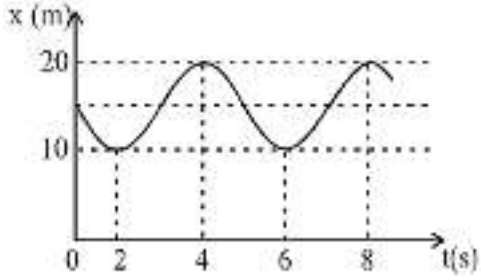
(1) A

(2) B

(3) both at same time

(4) need more information

2) Figure shows the position of a particle moving on the x-axis as a function of time. Choose the correct statement :-



(a) The particle has come to rest 4 times
(b) The maximum speed is at t = 4s
(c) The average velocity is zero for t = 2s to t=6s
(d) Motion of particle is non-uniformly accelerated

(1) a & b

(2) b & c

(3) only a & c

(4) a, c and d

3) A particle has initial velocity $4\hat{i} + 4\hat{j}$ m/s and an acceleration of $-0.4\hat{i}$ m/s^2. At what time will its speed be 5 m/s ?

(1) 2.5 sec

(2) 17.5 sec

(3) Both (1) and (2)

(4) None of these

4) A particle moves with an acceleration $a = k\sqrt{t}$. The average acceleration of the particle during a time interval from t = 0 to time t is

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(1) $\frac{a}{2}$

(2) $\frac{a}{3}$

(3) $\frac{2a}{3}$

(4) Zero

5) From a point on the ground at a distance 'a' from foot of a pole, a ball is thrown at an angle of 45° with horizontal which just touches the top of the pole and strikes the ground at a distance 'b' on other side. The height of pole will be :-

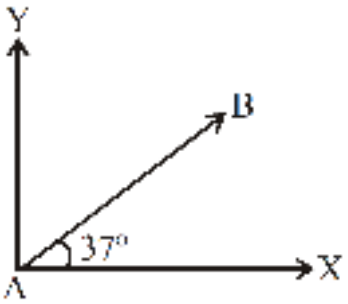
(1) $\frac{a+b}{2}$

(2) $\frac{ab}{a+b}$

(3) a+b

(4) $\frac{a+b}{ab}$

6) A butterfly is flying with velocity $(10\hat{i} + 12\hat{j})$ m/s with respect to wind and wind is blowing along x-axis with velocity u. If butterfly starts motion from A and after some time reaches point B. Then find the value of u:



(1) 16 m/s

(2) 8 m/s

(3) 6 m/s

(4) 12 m/s

7) A car starts moving along a line, first with acceleration a = 5 m/s^2 starting from rest, then uniformly and finally deaccelerating at the same rate 'a' comes to rest. The total time of motion is 25s. The average speed during the time is 20 m/s. How long does the particle move uniformly ?

(1) 10s

(2) 12s

(3) 20s

(4) 15s

8) A cyclist starts from rest and moves with a constant acceleration of 1 m/s^2. A boy who is 48 m behind the cyclist starts moving with a constant velocity of 10 m/s. After how much time the boy meets the cyclist ?

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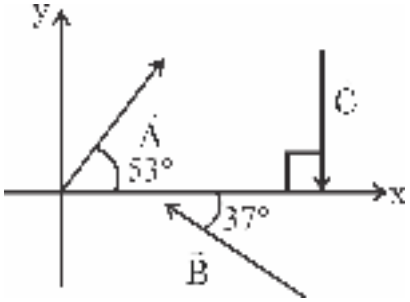
(1) 8 s

(2) 12 s

(3) 10 s

(4) Both (1) & (2)

9) Three vectors $\vec{A}, \vec{B}$ and $\vec{C}$ each of magnitude 10 units are shown in figure. Find resultant vector :



(1) $-2\hat{i}-8\hat{j}$

(2) $-2\hat{i}+4\hat{j}$

(3) $14\hat{i}+4\hat{j}$

(4) $14\hat{i}+24\hat{j}$

10) The vector $\hat{i} + x\hat{j} + 3\hat{k}$ is rotated through an angle θ and doubled in magnitude, then it becomes $4\hat{i} + (4x - 2)\hat{j} + 2\hat{k}$. The values of x is $\frac{n}{3}$. Then the value of 'n' is.

(1) 1

(2) –2

(3) 3

(4) –4

11) Two vectors $\vec{a}$ and $\vec{b}$ have magnitudes of 3 units and 4 units, respectively. Angle between $\vec{a}$ and $\vec{b}$ is θ

List-I		List-II	
(I)	$\vec{a}\cdot\vec{b}=0$	(P)	$\theta = 0$
(II)	$\hat{a}\cdot\vec{b}=4$ units	(Q)	$\theta = \pi$
(III)	$\vec{a}\cdot\vec{b}=-12$ units	(R)	$\theta=\frac{\pi}{4}$
(IV)	$\lvert\vec{a}+\vec{b}\rvert=5$ units	(S)	$\theta=\frac{\pi}{2}$
		(T)	$\theta=\frac{\pi}{3}$

(1) I-S, II-P, III-Q, IV-R

(2) I-S, II-P, III-Q, IV-S

(3) I-S, II-S, III-P, IV-S

(4) I-P, II-P, III-Q, IV-R

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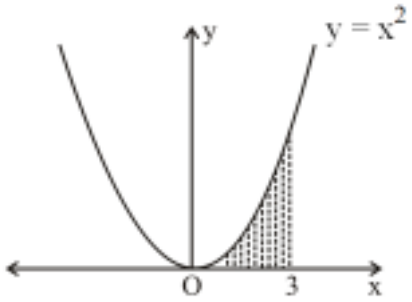
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12) The value of $\frac{1}{\pi}\int_0^{2\pi}(\sin^2 x)dx$ is

(1) 1
(2) 2
(3) 3
(4) 4

13)

In the diagram shown find the area of shaded region.



(1) 4
(2) 2
(3) 3
(4) 9

14) A river, while flowing, sweeps along with it few boulders (stones). Assume that the maximum mass m, of a boulder swept along by a river, depends on the speed v of the river, the acceleration due to gravity g and the density ρ of the boulder. If $m = kv^a g^b \rho^c$ where k is some dimensionless constant, the value of (a + b + c) is _____.

(1) 1
(2) 2
(3) 3
(4) 4

15) In a given system of unit standard measurement of mass is 100 gm, standard measurement of length is 200 cm and standard measurement of time is 5 sec. 10 J of energy in the given system of unit has value N, then value of N is.

(1) 200
(2) 225
(3) 375
(4) 625

16) The loss of pressure when a fluid flows through a pipe is given by $P = k \rho^a \square V^b d^c \mu$ where d and □ are diameter and length of the pipe respectively, ρ and μ are the mass density and coefficient of viscosity of the fluid, V is the mean velocity of flow through the pipe and k is a numerical constant.

Write the value of (a + b – c).

(1) 2
(2) 4
(3) 3
(4) –1

17) A physical quantity P is given as $P = \frac{a^2b^3}{c\sqrt{d}}$ The percentage error in the measurement of a, b, c and d are 1%, 2%, 3% and 4% respectively. The percentage error in the measurement of quantity P will be

(1) 13%
(2) 14%
(3) 12%
(4) 16%

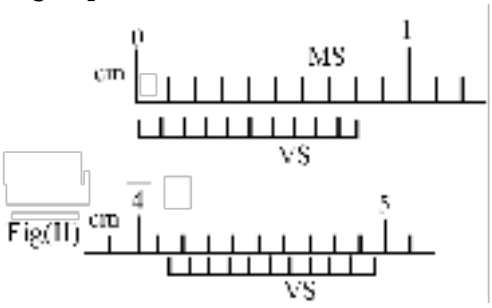
18) A simple pendulum is being used to determine the value of gravitational acceleration g at a certain place. The length of the pendulum is 25.0 cm and a stop watch with least count 1s measures the time taken for 40 oscillations to be 50 s. The percentage error in measurement of g is :

(1) 3.40%
(2) 5.40%
(3) 4.40%
(4) 2.40%

19) If 'e' denotes electrical charge 'ε_0' electrical permittivity of free space, 'h' Plancks constant and 'c' speed of light, then dimensions of $\left[\frac{e^2}{\varepsilon_0 hc}\right]$ are same as that of:-

(1) Angular momentum
(2) Energy
(3) Force
(4) Dimensionless

20) In a vernier calliper, when the jaws are closed the two scales are as shown in figure (I). When a ring is placed to measure its outer diameter, the position of two scales is as shown in figure (II).



(1) 4.11 cm

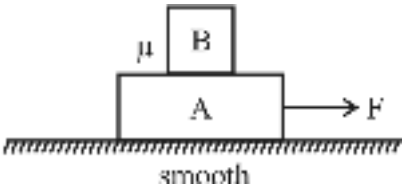
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(2) 4.12 cm

(3) 4.16 cm

(4) 4.24 cm

21) For the system shown in figure both the block A and B are moving together. If coefficient of friction between both the blocks is μ, the work done by friction on block B is :-



(1) Zero

(2) Negative

(3) Positive

(4) None of these

22) A drop of rain water of mass 2 g falling from a height of 0.5 km hits the ground with speed 30 m/s. Then work done by resistive force of air is

(1) –9.10 J

(2) –8.75 J

(3) +0.10 J

(4) –8.25 J

23) When a spring is stretched by a distance 'x', it exerts a force given by $F = -5x - 16x^3$ N. Work done by external force, when the spring is stretched from 0.1 m to 0.2 m, is :

(1) 12.2×10^{-1} J

(2) 12.2×10^{-2} J

(3) 8.7×10^{-1} J

(4) 8.1×10^{-2} J

24) The momentum of a body is increased by 25%. The kinetic energy is increased by about :

(1) 25%

(2) 5%

(3) 56%

(4) 38%

25) A body is initially at rest. It undergoes one-dimensional motion with constant acceleration. The power delivered to it at time t is proportional to-

(1) $t^{1/2}$

(2) t

(3) $t^{3/2}$

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(4) t^2

26) Calculate the maximum acceleration (in ms^{-2}) of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is 0.15 (g = $10ms^{-2}$).

(1) 50
(2) 1.2
(3) 150
(4) 1.5

27) Given below are two statements :
Statement-I : An elevator can go up or down with uniform speed when its weight is balanced with the tension of its cable.
Statement-II : Force exerted by the floor of an elevator on the foot of a person standing on it is more than his/her weight when the elevator goes down with increasing speed.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both statement I and statement II are false
(2) Statement I is true but statement II is false
(3) Both statement I and statement II are true
(4) Statement I is false but statement II is true

28) A machine gun fires 20 bullets per second into a target. Each bullet weighs 150 gms and has a speed of 800m/sec. Find the force necessary to hold the gun in position ..... N.

(1) 800
(2) 1000
(3) 1200
(4) 2400

29) If a car covers $\frac{2}{5}$ of the total distance with 2 m/s speed and $\frac{3}{5}$ distance with 3 m/s speed then average speed is

(1) 5 m/s
(2) 6 m/s
(3) 2.5 m/s
(4) 3.5 m/s

30) At any instant of time acceleration and velocity of particle are given by $\vec{a} = \hat{i} + \hat{j}$ & $\vec{v} = 6\hat{i} + 8\hat{j}$ then rate of change of speed at the same instant will be :-

(1) $\frac{(3\hat{i} + 4\hat{j})}{25}$
(2) $\frac{6\hat{i} + 8\hat{j}}{25}$

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(3) $\frac{7}{25}(3\hat{i}+8\hat{j})$

(4) $\frac{7}{25}(3\hat{i}+4\hat{j})$

31) A body thrown vertically upwards so as to reach its maximum height in t second. The total time from the time of projection to reach a point at half of its maximum height while returning (in second) is :

(1) $\sqrt{2}\,t$

(2) $\frac{t}{\sqrt{2}}$

(3) $t\left(1+\frac{1}{\sqrt{2}}\right)$

(4) $\frac{3t}{2}$

32) The equation of a projectile is $y=\sqrt{3}x-\frac{gx^2}{2}$. Find the speed of projectile at the maximum height of path.

(1) 2 m/s

(2) 1 m/s

(3) 3 m/s

(4) 4 m/s

33) If position vectors of particle A and particle B are $\vec{S}_A=(3t\hat{i}-4t^2\hat{j})m$ and $\vec{S}_B=(2t^2\hat{i}+t\hat{j})m$ then velocity of particle B as seen by particle A at t = 1 sec. is.

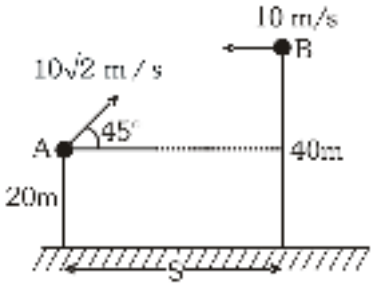
(1) $(\hat{i}+9\hat{j})$

(2) $-(\hat{i}+9\hat{j})$

(3) $3\hat{i}-8\hat{j}$

(4) $4\hat{i}+\hat{j}$

34) Two projectile are projected from the top of the two tower simultaneously as shown. If they collide in air, then the distance 'S' between towers is :



(1) 40 m

(2) 20 m

(3) $20\sqrt{2}$ m

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(4) 30 m

35) A man can swim in still water at a speed of 5 km/hr. He wants to cross a river 6 km wide, flowing at the rate of 4 km/hr. If he heads in a direction making an angle of 127° with stream direction, then he will reach a point on the other bank :

(1) upstream at a distance of 1.5 km

(2) downstream at a distance of 1.5 km

(3) directly on the other side of the bank

(4) never reach the other bank

36) A 30 kg block rests on rough horizontal surface. A horizontal force of 200 N is applied on the body. The block acquires a speed of 4 m/s, starting from rest, in 2s. What is the value of coefficient of friction? ($g = 10\ m/s^2$)

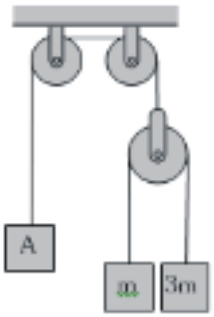
(1) $10\sqrt{3}$

(2) $\frac{\sqrt{3}}{10}$

(3) 0.19

(4) 0.47

37) In the given figure, find mass of the block A, if it remains at rest, when the system is released



from rest. Pulleys and strings are ideal.

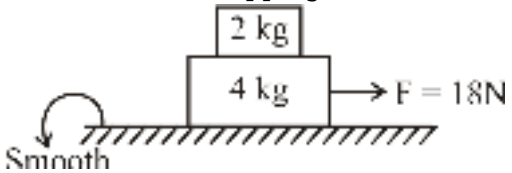
(1) m

(2) 2m

(3) 2.5m

(4) 3m

38) There is no slipping between the two blocks. What is force of friction between two block ?



(1) Zero

(2) 9N

(3) 12N

(4) 6N

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39) A rocket of initial mass 1500 kg ejects gas at a constant rate of 10 kg/s with a relative speed of 5 km/s. What is the acceleration of the rocket 50 seconds after the launch, neglecting gravity:-

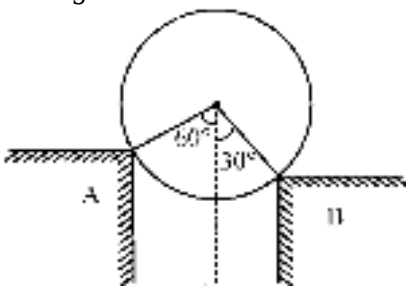
(1) 10 m/s^2

(2) 25 m/s^2

(3) 50 m/s^2

(4) 100 m/s^2

40) A smooth cylinder of mass m and radius R is resting on two corner edges A and B as shown in fig. The relation between normal reaction at the edges A and B is :-



(1) $N_A = \sqrt{2}N_B$

(2) $N_B = \frac{2\sqrt{3}N_A}{5}$

(3) $N_A = \frac{N_B}{2}$

(4) $N_B = \sqrt{3}N_A$

41) A ball is suspended on a thread from the ceiling of a car. The brakes are applied and the speed of car changes from 5 m/sec to $\frac{5}{3}$ m/sec during the time interval of 3 seconds. The angle that the thread will deviate from vertical.

(1) $\theta = \tan^{-1}\left(\frac{1}{9}\right)$

(2) $\theta = \tan^{-1}\left(\frac{8}{9}\right)$

(3) $\theta = \sin^{-1}\left(\frac{1}{9}\right)$

(4) $\theta = \cos^{-1}\left(\frac{1}{9}\right)$

42) A machine rated as 150W, changes the velocity of a 10kg mass from 4m/s to 10m/s in 4s. The efficiency of the machine is nearly.

(1) 70%

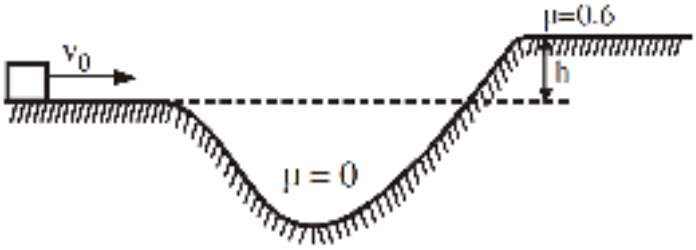
(2) 30%

(3) 50%

(4) 40%

43) In the figure, a block slides along a track from one level to a higher level, by moving through an

intermediate valley. The track is frictionless untill the block reaches the higher level. There a frictional force stops the block in a distance d. The block's initial speed v_0 is 6 m/s, the height difference h is 1.1m and the coefficient of kinetic friction μ is 0.6. The value of d is



(1) 1.17 m

(2) 1.71 m

(3) 7.11 m

(4) 11.7 m

44)

A ball is released from the top of a tower. The ratio of work done by force of gravity in first, second and third second of the motion of the ball is :

(1) 1 : 2 : 3

(2) 1 : 4 : 9

(3) 1 : 3 : 5

(4) 1 : 5 : 3

45) A bullet is fired at a target with a velocity 80 m/s and penetrates 50 cm into it. If this bullet were fired into a target 25 cm thick with equal velocity, with what velocity would it emerge, supposing the resistance to be uniform and same in both the cases-

(1) $\sqrt{80}$ m/s

(2) $\frac{40}{\sqrt{2}}$ m/s

(3) 40 m/s

(4) $40 \times \sqrt{2}$ m/s

CHEMISTRY

1) The correctly reported difference of 16.4215 and 6.01 will have significant figures equal to:-

(1) 3

(2) 4

(3) 5

(4) 6

2) Calculate how many methane molecules & how many carbon and hydrogen atoms respectively are there in 25.0 g of methane ?

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(1) 9.41×10^{23}, 9.41×10^{23}, 37.64×10^{23}

(2) 9.41×10^{23}, 37.64×10^{20}, 2.4×10^{22}

(3) 9.41×10^{23}, 2.37×10^{23}, 37.64×10^{23}

(4) None of the above

3) 1.5 g of NaI reacts with 1.6 g of Br_2 to produce 0.412 g of NaBr, then determine the percentage yield of the reaction ?
[Atomic weight of Na = 23, Br = 80, I = 127]
$2NaI + Br_2 \rightarrow 2NaBr + I_2$

(1) 30%

(2) 40%

(3) 50%

(4) 60%

4) Calculate weight of 50% pure $CaCO_3$ required to obtain 11 g CO_2 :-

(1) 12.5 g

(2) 30 g

(3) 50 g

(4) 75 g

5) How many moles of lead (II) chloride will be formed from a reaction between 6.5 g of PbO and 3.2 g of HCl ? (Atomic weight of Pb = 207) $PbO + 2\ HCl \rightarrow PbCl_2 + H_2O$

(1) 0.011

(2) 0.029

(3) 0.044

(4) 0.333

6) Which of the following statement are correct?
(A) The constituents of a compound cannot be separated into simpler substances by physical methods.
(B) The properties of a compound are always similar to properties of its constituent elements.
(C) S.I. unit of volume is litre (L).
(D) S.I. unit of pressure is bar.
(E) Freezing point and boiling point of water is 32°F and 212°F respectively.

(1) A, B and E

(2) B, C and E

(3) A, C and D

(4) A and E only

7) Rearrange the following (I to IV) in the order of increasing masses :
(I) 0.5 mole of O_3
(II) 0.5 gram atom of oxygen
(III) 3.011×10^{23} molecules of O_2

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(IV) 5.6 litre of CO_2 gas at STP

(1) II < IV < III < I
(2) II < I < IV < III
(3) IV < II < III < I
(4) I < II < III < IV

8) On analysis, a certain compound was found to contain 254 g of iodine (atomic mass 127) and 80 g oxygen (atomic mass 16). What is the formula of the compound ?

(1) IO
(2) I_2O
(3) I_5O_3
(4) I_2O_5

9) Which of the following statements are not true for the reaction : $4Fe + 3O_2 \rightarrow 4Fe^{+3} + 6O^{-2}$
(a) It is a redox reaction
(b) Metallic iron is a reductant
(c) Fe^{+3} is a reductant
(d) Metallic iron is reduced to Fe^{+3}
Correct option is :

(1) c, d
(2) a, b
(3) a, c
(4) a, b, d

10) $Zn + NO_3^- \rightarrow Zn^{+2} + NH_4^+$
The coefficient of NO_3^-, Zn and H^+ in balanced equation respectively are :-

(1) 4, 1, 10
(2) 10, 4, 1
(3) 1, 4, 10
(4) 4, 1, 7

11) **Assertion :–** Nitrous acid (HNO_2) may act as an oxidising agent as well as a reducing agent.
Reason :– The oxidation number of nitrogen remains same in all the compounds.

(1) Both Assertion & Reason are True & Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but Reason is False.
(4) Both Assertion & Reason are False.

12) Match the following processes with number of electrons transferred per mole :-

	Column-I		Column-II

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(A)	$Fe(NO_3)_3 \rightarrow Fe(NO_2)_2$	(P)	6
(B)	$Cu_2S \rightarrow Cu^{2+} + SO_2$	(Q)	4
(C)	$N_2H_4 \rightarrow N_2$	(R)	7
(D)	$KI \rightarrow KIO_3$	(S)	8

(1) A → P; B → S; C → Q; D → R

(2) A → Q; B → R; C → P; D → S

(3) A → S; B → P; C → Q; D → R

(4) A → R; B → S; C → Q; D → P

13) Match the following reactions with their type :

	Column-I		**Column-II**
(A)	$2KClO_3 \rightarrow 2KCl + 3O_2$	(P)	Non-redox
(B)	$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$	(Q)	Disproportionation
(C)	$PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$	(R)	Intra molecular redox
(D)	$P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$	(S)	Inter-molecular redox

(1) A → S; B → P; C → Q; D → R

(2) A → Q; B → S; C → R; D → P

(3) A → R; B → P; C → S; D → Q

(4) A → R; B → Q; C → P; D → S

14) ***Assertion :-*** The value of equilibrium constant (K) increases when concentration of the reactants are increased.
Reason :- On increasing concentration of gaseous reactants, equilibrium shifts in forward direction.

(1) Both assertion and reason are true and reason is correct explanation of assertion.

(2) Both assertion and reason are true but reason is not a correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

15) **Statement I :** The solubility of gases in liquid always increases with decrease in pressure.
Statement II : According to Le-Chatelier's principle high pressure favours the change in reaction where number of gaseous moles decreases.

(1) Statement-I is false and Statement-II is true.

(2) Both Statement-I & Statement-II are false.

(3) Both Statement-I and Statement-II are true and Statement-II is the correct explanation of Statement-I.

(4) Both Statement-I & Statement-II are true but Statement-II is not a correct explanation of Statement-I.

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16) Match Column–I (hypothetical reactions) with Column–II (ratio $\frac{K_P}{K_C}$ for the given reactions) and select the correrct answer using the code given below the lists :-

	Column–I		**Column–II**
(A)	$A_2(g) + 3B_2(g) \rightleftharpoons 2AB_3(g)$	(P)	$(RT)^{-2}$
(B)	$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$	(Q)	$(RT)^{0}$
(C)	$A(s) + 1.5\ B_2(g) \rightleftharpoons AB_3(g)$	(R)	$(RT)^{\frac{1}{2}}$
(D)	$AB_2(g) \rightleftharpoons AB(g) + 0.5\ B_2(g)$	(S)	$(RT)^{-\frac{1}{2}}$

(1) (A)–(S), (B)–(R), (C)–(Q), (D)–(P)

(2) (A)–(P), (B)–(S), (C)–(R), (D)–(Q)

(3) (A)–(P), (B)–(Q), (C)–(S), (D)–(R)

(4) (A)–(R), (B)–(P), (C)–(Q), (D)–(S)

17) **Assertion :** pH of solution of CH_3COOH decreases on addition of sodium acetate.
Reason : Sodium acetate suppresses the degree of dissociation of CH_3COOH by common ion effect.

(1) Both Assertion & Reason are True and Reason is the correct explanation of the Assertion

(2) Both Assertion & Reason are True but Reason is not the correct explanation of the Assertion

(3) Assertion is True but Reason is false

(4) Assertion is False but Reason is True

18) Which of the following mixture (aqueous) will act as a buffer solution?

(1) 0.1 mole CH_3COOH + 0.1 mole HCOOH

(2) 0.1 mole HCOOH + 0.2 mole HCOONa

(3) 0.1 mole NaOH + 0.1 mole NaCl

(4) 0.1 mole NH_4OH + 0.2 mole C_6H_5–NH_2

19) How many of the following compound show cationic-anionic hydrolysis ?
NH_4Cl, KCl, NaCl, NH_4CN, $HCOONH_4$, $(NH_4)_2SO_4$, K_2SO_4

(1) 2

(2) 3

(3) 4

(4) 6

20) Let the solubilities of AgCl in pure water, in 0.1 M NaCl solution and in 0.1 M NH_3 solution be s_1, s_2 and s_3 respectively. Then choose correct option :-

(1) $s_1 > s_2 > s_3$

(2) $s_3 > s_2 > s_1$

(3) $s_2 > s_1 > s_3$

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(4) $s_3 > s_1 > s_2$

21) At 25°C in following for which pair $pK_a + pK_b = 14$:-
(a) H_2SO_4, SO_4^{-2} (b) HCl, HSO_4^-
(c) H_2SO_3, SO_3^{-2} (d) H_2CO_3, HCO_3^-

(1) a, b, d
(2) b, d
(3) only d
(4) c, d

22) If at a certain temperature, K_w (H_2O) = 10^{-10}. A solution of pH = 5.4, under this condition will be :

(1) Basic
(2) Acidic
(3) Neutral
(4) Amphoteric

23) In the reaction $H_2S \rightleftharpoons 2H^+ + S^{-2}$. When NH_4OH is added then :–

(1) S^{-2} is precipitate
(2) No action takes place
(3) Concentration of S^{-2} decreases
(4) Concentration of S^{-2} increases

24) Match List I with List II and choose the correct answer from the code :

	List I Electrolyte		**List II Solubility product**
(a)	Bi_2S_3	(i)	$4s^3$
(b)	CdS	(ii)	$27s^4$
(c)	$Al(OH)_3$	(iii)	$108s^5$
(d)	CaF_2	(iv)	s^2

Code : (a) (b) (c) (d)
(1) (iii) (iv) (ii) (i)
(2) (i) (ii) (iii) (iv)
(3) (iv) (i) (ii) (iii) (4) (ii) (iii) (i) (iv)

(1) 1
(2) 2
(3) 3
(4) 4

25) Which has highest pH ?

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(1) $NaCl_{(aq)}$

(2) $NH_4Cl_{(aq)}$

(3) $Na_2CO_{3(aq)}$

(4) $CuSO_{4(aq)}$

26) The enthalpy change of the reaction,
$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$ is ΔH_1 and that of $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$ is ΔH_2 then–

(1) $|\Delta H_1| > |\Delta H_2|$

(2) $\Delta H_1 + \Delta H_2 = 0$

(3) $|\Delta H_1| < |\Delta H_2|$

(4) $\Delta H_1 = \Delta H_2$

27) Enthalpies of formation of CO(g), CO_2(g), N_2O(g) and N_2O_4(g) are –110, –393, 81 and 9.7 kJ mol^{-1} respectively. Find the value of ΔH for the reaction :
$N_2O_4(g) + 3CO(g) \rightarrow N_2O(g) + 3CO_2(g)$

(1) +778 kJ mol^{-1}

(2) –211.7 kJ mol^{-1}

(3) +211.7 kJ mol^{-1}

(4) –778 kJ mol^{-1}

28) Enthalpy of neutralization of HCl by NaOH is –55.84 kJ/mol and by NH_4OH is –51.34 kJ/mol. The enthalpy of ionization of NH_4OH is

(1) –107.18 kJ/mol

(2) 107.18 kJ/mol

(3) 4.5 kJ/mol

(4) –4.5 kJ/mol

29)

If $S + O_2 \rightarrow SO_2$; $\Delta H = -298.2$ kJ

$SO_2 + \frac{1}{2}O_2 \rightarrow SO_3$; $\Delta H = -98.7$ kJ

$SO_3 + H_2O \rightarrow H_2SO_4$; $\Delta H = -130.2$ kJ

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$; $\Delta H = -227.3$ kJ

the heat of formation of H_2SO_4 will be :-

(1) –754.4 kJ

(2) +320.5 kJ

(3) –650.3 kJ

(4) –433.7 kJ

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30)

The enthalpy change for the reaction
$H_2(g) + C_2H_4(g) \rightarrow C_2H_6(g)$ is..... The bond energies are,
H–H ⇒ 110, C—H ⇒ 100, C—C ⇒ 80 & C=C ⇒ 150 Kcal mol^{-1}

(1) –10 Kcal mol^{-1}
(2) +10 Kcal mol^{-1}
(3) – 20 Kcal mol^{-1}
(4) +30 Kcal mol^{-1}

31) For a gaseous reaction $A_{(g)} + 3B_{(g)} \rightarrow 3C_{(g)} + 3D_{(g)}$ ΔE is 17 kcal at 27°C.
Assuming R = 2 cal K^{-1} mol^{-1}, the value of ΔH for the above reaction will be:

(1) 15.8 kcal
(2) 16.4 kcal
(3) 18.2 kcal
(4) 20.0 kcal

32) Consider the following reaction
$C(s) + O_2(g) \rightarrow CO_2(g) + x$ kJ
$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g) + y$ kJ
The heat of formation of CO(g) is :

(1) –(x + y)kJ/mol
(2) (x – y)kJ/mol
(3) (y – x)kJ/mol
(4) None of these

33) For the reaction : $N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$; ΔH = 30 kJ mol^{-1},
ΔS = 75 J K^{-1} mol^{-1}. The temperature at which above reaction is spontaneous?

(1) 400 K
(2) 300 K
(3) 150 K
(4) 425 K

34) Match the following columns :-

	Column-I		**Column-II**
(A)	$CH_4(g) \rightarrow C(g) + 4H(g)$	(P)	Enthalpy of solution
(B)	$6C(graphite) + 3H_2(g) \rightarrow C_6H_6(l)$	(Q)	Enthalpy of combustion
(C)	$C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$	(R)	Lattice enthalpy
(D)	$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$	(S)	Enthalpy of atomization

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(E)	$NaCl(s) \rightarrow Na^+(g) + Cl^-(g)$	(T)	Enthalpy of formation
(F)	$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$	(U)	Enthalpy of hydrogenation

Correct match will be :-

(1) A-R, B-S, C-P, D-Q, E-U, F-T

(2) A-U, B-T, C-S, D-R, E-P, F-Q

(3) A-S, B-T, C-Q, D-P, E-R, F-U

(4) A-S, B-T, C-U, D-R, E-P, F-Q

35)

Standard entropy of X_2, Y_2 and XY_3 are 60, 40 and 50 $JK^{-1}mol^{-1}$. For the reaction, $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$, If $\Delta H = -30$ KJ, to be at equilibrium the temperature will be :-

(1) 1250 K

(2) 500 K

(3) 750 K

(4) 1000 K

36)

For which reaction, ΔS will be maximum ?

(1) $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

(2) $N_2(g) + 2O_2(g) \rightarrow N_2O_4(g)$

(3) $S(s) + \frac{3}{2} O_2(g) \rightarrow SO_3(g)$

(4) $2C(s) + O_2(g) \rightarrow 2CO(g)$

37) Select Incorrect statement :-

(1) An adiabatic system can exchange energy with its surroundings.

(2) A thermodynamic property which is intensive is additive.

(3) Work done may be zero in a cyclic process

(4) For a simple compressible substance, the relation $dq - P.dV = 0$ is true for any cycle process involving mechanical work only

38) The value of enthalpy change (ΔH) for the reaction
$C_2H_5OH_{(\square)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 3H_2O_{(\square)}$
at 27°C is –1366.5 kJ mol^{-1}. The value of internal energy change for the above reaction at this temperature will be :-

(1) –1371.5 kJ

(2) –1369.0 kJ

(3) –1364.0 kJ

(4) –1361.5 kJ

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39) For process, $H_2O(\square) \rightleftharpoons H_2O(s)$ at 273 K and 1 atm
Select the correct option.

(1) ΔH = – ve, ΔS = +ve, ΔV = –ve, ΔG = 0
(2) ΔH = –ve, ΔS = –ve, ΔV = +ve, ΔG = 0
(3) ΔH = +ve ΔS = +ve, ΔV = –ve, ΔG = 0
(4) ΔH = –ve, ΔS = –ve, ΔV = –ve, ΔG = 0

40) Which of the following statement is correct :-

(1) In isobaric process heat absorbed by system is equal to decrease in enthalpy of system.
(2) In isochoric process heat released by system is equal to increase in internal energy of system.
(3) In adiabatic process work done by system is equal to decrease in internal energy of system.
(4) In cyclic process work done on system is equal to the heat absorbed by system.

41) In an isothermal expansion of an ideal gas, the work done is equal to-

(1) Heat released out of the system
(2) Heat absorbed by the system
(3) Zero
(4) None of them

42) During adiabatic expansion of an ideal gas in vacuum :-

(1) q = 0 and $\Delta U \neq 0$
(2) q ≠0 and $\Delta U \neq 0$
(3) q ≠ 0 and $\Delta U = 0$
(4) q = 0 and $\Delta U = 0$

43) Which of the following statement is incorrect?

(1) At equilibrium, $(\Delta G)_{system}$ is equal to zero
(2) Endothermic processes are never spontaneous
(3) For a spontaneous process, $(\Delta S)_{total}$ is greater than zero
(4) Any extensive property is always intensive if it is defined for 1 mol.

44) ΔU equal to :

(1) Isobaric work
(2) Adiabatic work
(3) Isothermal work
(4) Isochoric work

45) **Assertion (A) :** Spontaneous process is an irreversible process and can be reversed by some external agency
Reason (R) : Decrease in enthalpy is a contributory factor for spontaneity.

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(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BIOLOGY

1) Which of the following statement is incorrect?

(1) Fungi are cosmopolitan in nature

(2) The cell walls of all fungi are composed of cellulose and polypeptides

(3) Sexual reproduction in fungi is followed by plasmogamy, karyogamy & meiosis.

(4) Some fungi show dikaryotic stage in sexual reproduction.

2) Select the incorrect statement-

(1) Diatoms and desmids float passively in water current

(2) Dianoflagellates and Euglenoids are chief produces of ocean

(3) In diatoms, cell wall form two thin overlapping shells which fit together as in a soapbox

(4) Dead diatoms cell wall form diatomaceous earth.

3) Read the following statements carefully.
(A) They have a protein rich layer called aleurone which make their body flexible.
(B) They have two flagella a short and a long one.
(C) Pigments of euglenoids are identical to those present in higher plants.
(D) They are photosynthetic in the presence of sunlight when deprived of sunlight they behave like heterotrophs.
How many above statements are correct regarding euglenoids.

(1) One

(2) Two

(3) Three

(4) Four

4) Match list-I with list-II

	List-I		**List-II**
(A)	Cell wall with stiff cellulosic plates	I.	Protozoans
(B)	Pigments identical to higher plants	II.	Diatoms
(C)	Chief producers in oceans	III.	Dinoflagellates
(D)	Primitive relatives of animals	IV.	Euglenoids

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Choose the correct answer from the options given below :-

(1) A-IV, B-III, C-I, D-II

(2) A-III, B-IV, C-I, D-II

(3) A-III, B-IV, C-II, D-I

(4) A-IV, B-III, C-II, D-I

5) Match the list-1 with list-2

	List-1		**List-2**
(A)	*Mucor*	(i)	golden algae
(B)	Desmid	(ii)	Imperfect fungi
(C)	*Paramoecium*	(iii)	Bread mould
(D)	*Trichoderma*	(iv)	Ciliated protozoans

(1) A-(ii), B-(iv), C-(i), D-(iii)

(2) A-(iii), B-(i), C-(iv), D-(ii)

(3) A-(iii), B-(ii), C-(iv), D-(i)

(4) A-(iv), B-(ii), C-(i), D-(iii)

6) **Assertion (A) :** Chemosynthetic autotropic bacteria oxidise various inorganic substances such as nitrates and ammonia.
Reason (R) : They (bacteria) are helpful in making curd from milk, Production of antibiotic, fixing of nitrogen in legumes.

(1) Both Assertion and Reason are correct and Reason is correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion and Reason both are false.

(4) Both Assertion and Reason are true, but Reason is not correct explanation of Assertion.

7) **Statement-I :-** The sexual spores of Deuteromycetes are conidia.
Statement-II :- Some members of deuteromycetes are saprophytes or parasites while a large number of them are decomposers of litter and help in mineral cycling.

(1) Statement-I and statement-II both are correct.

(2) Statement-I and statement-II both are incorrect.

(3) Statement-I is correct, statement-II is incorrect.

(4) Statement-I is incorrect, statement-II is correct.

8) The asexual spores of Deuteromycetes are _____(A)_____ and members of this fungal class are called _____(B)____ .
Select the option which correctly fill the blanks (A) & (B) :-

(1) (A) - Zoospore, (B) - Algal fungi

(2) (A) - Basidiospore, (B) - Club fungi

(3) (A) - Conidia, (B) - Imperfect fungi

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(4) (A) - Aplanospores, (B) - Sac fungi

9) How many of the following belongs to sac fungi ?
Penicillium, A.niger, Claviceps, Morchella, Neurospora, Yeast, Agaricus, Ustilago and *Mushroom.*

(1) Five
(2) Six
(3) Seven
(4) Eight

10) Read the following statements about members of Gymnosperms.
A. The roots are generally tap roots
B. The leaves in gymnosperms are well-adapted to withstand extremes of temperature, humidity and wind
C. The leaves are always compound
D. The ovules are not enclosed by ovary wall and remain exposed, both before and after fertilisation.
How many of the above statements are correct :-

(1) Four
(2) Three
(3) Two
(4) One

11) Members of Pteridophyta -
A. They produce spores by mitosis in spore mother cells.
B. They form spores which germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes called prothallus.
C. They perform fusion of male gamete with the megaspore present in the archegonium result in the formation of zygote -

(1) Only A is incorrect.
(2) Only B is incorrect.
(3) Only C is incorrect.
(4) Both A and C are incorrect.

12) Choose the incorrect statement :-

(1) Most of the members of chlorophyceae have one or more storage bodies called pyrenoids.
(2) Members of rhodophyceae reproduce asexually by biflagellated zoospore.
(3) In members of phaeophyceae union of gametes may take place in water or within the oogonium
(4) Chlorophyll 'a' is common in all class of algae of kingdom plantae.

13) Match column-I, II & III and choose correct option :

	Column-I		**Column-II**		**Column-III**
A	Chlorophyceae	P	Chlorophyll a, d	I	Cyanophycean starch

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B	Cyanobacteria	Q	Chlorophyll a, b	II	Starch
C	Phaeophyceae	R	Chlorophyll a, c	III	Mannitol
D	Rhodophyceae	S	C-Phycocyanin	IV	Floridean starch

(1) A-Q-II, B-R-III, C-S-I, D-P-IV

(2) A-S-I, B-R-III, C-P-IV, D-Q-II

(3) A-S-I, B-Q-II, C-P-IV, D-R-III

(4) A-Q-II, B-S-I, C-R-III, D-P-IV

14) Match the following columns and choose the correct option :

Column-I		**Column-II**	
(A)	Flagellated isogametes	(i)	*Spirogyra*
(B)	Non flagellated isogametes	(ii)	*Ulothrix*
(C)	Anisogamy	(iii)	*Eudorina*
(D)	Oogamy	(iv)	*Fucus*

(1) (A)-(i), (B)-(ii), (C)-(iv), (D)-(iii)

(2) (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)

(3) (A)-(ii), (B)-(i), (C)-(iii), (D)-(iv)

(4) (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)

15) Given below are two statements : One is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A) :- Asexual reproduction in liverworts take place by fragmentation of thalli; or by the formation of specialised structure called gemmae.
Reason (R) :- Gammae are green, multicellular, sexual buds which develop in small receptacles called gemma cup located on the thalli.

(1) Both A & R are true but R is not the correct explanation of A.

(2) A is true but R is false.

(3) A is false but R is true.

(4) Both A & R are true and R is correct explanation of A.

16) Given below are two statements : One is labelled as assertion (A) and the other is labelled as reason (R) :
Assertion (A) :- Plant body of bryophytes is more differentiated than that of algae.
Reason (R) :- Bryophyte **always** possess root-like, stem-like and leaf-like structure in its main plant body.
Choose the correct answer from the options given below :-

(1) Both (A) and (R) are true but (R) is not correct explanation of (A).

(2) (A) is false but (R) is true.

(3) Both (A) and (R) are true and (R) is correct explanation of (A).

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(4) (A) is true but (R) is false.

17) Read both statements carefully.
Statement-I : The sporophylls may form distinct compact structure called strobili in some pteridophytes.
Statement-II : Gametophyte of pteridophytes is thalloid, mostly nonphotosynthetic but multicellular.
Select the correct option.

(1) Statement-I is incorrect but statement-II is correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Both statement-I and statement-II are correct.
(4) Both statement-I and statement-II are incorrect.

18) **Statement-I :** The male gametophytic generation is highly reduced in gymnosperms.
Statement-II : In gymnosperms, the male gametophytes do not have an independent free-living existence.

(1) Statement-I is true but Statement-II is false.
(2) Both Statement-I and Statement-II are true.
(3) Statement-I is false but Statement-II is true.
(4) Both Statement-I and Statement-II are false.

19) Biological names are generally in ___A___ and written in ___B___ .
Fill in the blanks A and B :

(1) A - Greek, B - Italics
(2) A - Latin, B - Italics
(3) A - English, B - Italics
(4) A - German, B - Italics

20) Both mango and wheat are placed together at which taxonomic category ?

(1) Class
(2) Order
(3) Family
(4) Division

21) How many term given below are related to methanogens :
Archaebacteria, Ruminant animals, Eukaryotic, Photosynthetic, Biogas

(1) Five
(2) Three
(3) Four
(4) Six

22) Taxonomic categories are distinct biological _____ and not merely morphological _____

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(1) aggregates, entities
(2) entities, aggregates
(3) entities, category
(4) aggregates, taxon

23) In taxonomic hierarchy, genus comes between __I__ and __II__.
Select correct option to fill I and II.

(1) I - Family, II - Class
(2) I - Order, II - Division
(3) I - Class, II - Kingdom
(4) I - Species, II - Family

24) Reproduction is synonymous with growth for which organisms -

(1) Bacteria, multicellular algae, hydra
(2) Bacteria, unicellular algae, *Amoeba*
(3) *Planaria*, unicellular algae, *Amoeba*
(4) Bacteria, fungi, *Amoeba*

25) The members of ascomycetes differ from basidiomycetes in having :

(1) Branched and septate mycelium.
(2) Conidia formation
(3) Formation of haploid sexual spores
(4) Formation of diploid sexual spores.

26) Mycorrhiza is :-

(1) A fungus parasiting root system of higher plants
(2) An association of *Rhizobium* with roots of leguminous plants
(3) A symbiotic association of plant roots and certain fungi
(4) An association of algae with fungi

27) An infectious agent that are smaller than viruses and causes potato spindle tuber disease, is :

(1) Virus
(2) Prion
(3) Viroid
(4) Bacteria

28) In which fungi asexual spores are produced exogenously while sexual spores are produced endogenously ?

(1) *Agaricus*
(2) *Puccinia*

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(3) *Aspergillus*

(4) *Ustilago*

29) Which of the following is not a criteria of five kingdom system ?

(1) Presence or absence of well-defined nucleus

(2) Stored food material

(3) Mode of nutrition

(4) Phylogenetic relationships

30) Which of the following belongs to class basidomycetes.

(i) *Agaricus* (ii) *Trichoderma*

(iii) *Puccinia* (iv) *Saccharomyces*

(v) *Ustilago*

(1) i and ii only

(2) ii and iii only

(3) iii, iv, v only

(4) i, iii and v only

31) According to Whittaker's classification how many kingdom includes cell walled multicellular organism?

(1) One

(2) Two

(3) Three

(4) Four

32) Small circular extrachromosomal DNA present in bacterial cell is known as :

(1) Mesosome

(2) Plasmid

(3) Chromatophore

(4) Genophore



33) Which of the following option is not correct about given diagram?

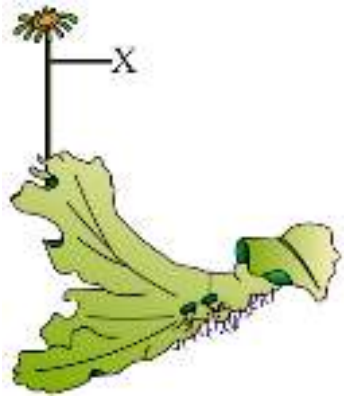
(1) The main plant is heterosporous.

(2) Both seeds and fruits are produced during sexual reproduction.

(3) Pollen-tube is formed.

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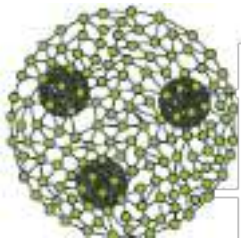
(4) The main plant body is sporophyte.



34) Given below is the diagram of *Marchantia,* labelled structure 'X' is :-

(1) Archegoniophore
(2) Antheridiophore
(3) Seta
(4) Capsule

35) Identify the diagram A, B and C :

(A) 

(B) 



(C)

	A	B	C
(1)	*Spirogyra*	*Fucus*	*Polysiphonia*
(2)	*Volvox*	*Fucus*	*Polysiphonia*
(3)	*Ulothrix*	*Fucus*	*Polysiphonia*
(4)	*Polysiphonia*	*Fucus*	*Ulothrix*

(1) 1
(2) 2
(3) 3

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(4) 4

36) Gymnosperms are :-

(1) Naked seeded vascular embryophytes
(2) Naked seeded vascular non-embryophytes
(3) Naked seeded non-vascular embryophytes
(4) Covered seeded vascular embryophytes

37) The sex organs in bryophytes are -

(1) Unicellular
(2) Multicellular
(3) Both 1 & 2
(4) Acellular

38) Which of the following structures is not associated with sporophyte of bryophytes.

(1) Foot, seta, capsule
(2) Spore mother cell
(3) Structures associated with Elaborate mechanism of spore dispersal
(4) Gemma cup

39) Observe the following spores and answer the given question :-
Zoospores, Conidia, Aplanospores, Endospores, Oospores, Zygospores, Hypnospores, Akinetes, Basidiospores.
How many of the above are formed during sexual reproduction in fungi :-

(1) Five
(2) Three
(3) Four
(4) Six

40) The female gametophytes are retained on parental sporophyte for variable time period in:

(1) *Equisetum*
(2) *Marchantia*
(3) *Selaginella*
(4) *Chara*

41) The correct sequence of events in the life cycle of bryophytes is :-
A : Production of antherozoids and egg
B : Fusion of gametes
C : Formation of sporophyte
D : Meiosis in spore mother cell
E : Germination of spores

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(1) A, D, B, C, E

(2) A, B, C, E, D

(3) A, B, C, D, E

(4) A, E, D, C, B

42) Choose the correct one from given statements.
A. Gametes in rhodophyceae are pyriform and bear two laterally attached flagella.
B. Sexual reproduction is oogamous and accompanied by complex post fertilisation developments in rhodophyceae.
C. The chloroplasts may be discoid, plate -like, reticulate, cup-shaped, spiral or ribbon- shaped in different species of Green algae.
D. Fusion between one large, non-motile, static female gamete and a smaller, motile male gamete is termed oogamous.
E. Oogamous reproduction occurs in *Volvox*, *Fucus*.

(1) (A) and (D)

(2) All except (A)

(3) All except (E)

(4) All of these

43) How many of the following statements are correct.
(i) Roots in *pinus* have fungal association in the form of mycorrhiza.
(ii) The roots are generally coralloid roots in gymnosperm.
(iii) Most of the gymnosperms are heterosporous plants and some are homosporous
(iv) Gymnosperms includes tall tree with always branched stem
(v) In *Cycas* male cone and female cones are born on different branches.

(1) Two

(2) One

(3) Zero

(4) Three

44) How many organisms in the list given below have cellulosic cell wall.
Methanogens, *Nostoc*, *Mycoplasma*, Diatoms, *Gonyaulax*, *Euglena*, *Trypanosoma*, Yeast, *Agaricus*, *Puccinia*, *Trichoderma*

(1) Six

(2) Three

(3) Four

(4) Two

45) Following are the taxonomic categories :-
A. Division
B. Order
C. Family
D. Genus
E. Species
F. Kingdom

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G. Class
Choose the correct sequence of taxonomic categories from the options given below :-

(1) E → D → B → C → G → A → F
(2) A → F → G → B → C → D → E
(3) F → A → B → G → C → D → E
(4) E → D → C → B → G → A → F

46) **Statement-I :** In male frog, urethra also called urino genital duct.
Statement-II : The urino genital duct in male frogs carries both sperm and urine.

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Only statement I is correct.
(4) Only statement II is correct.

47) **Assertion :** Frog respires through skin.
Reason : Frog have mucous glands in its skin and skin is highly vasculized.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

48) Match the column in refrence to frog :-

Column-I		**Column-II**	
A.	Cutaneous respiration	(i)	receive sound signals
B.	Tympanum	(ii)	1st digit of fore-limbs
C.	Copulatory pad	(iii)	through skin
D.	Ureotelic	(iv)	Excrete urea

(1) A–i, B–ii, C–iii, D–iv
(2) A–ii, B–i, C–iii, D–iv
(3) A–iii, B–i, C–ii, D–iv
(4) A–iii, B–i, C–iv, D–ii

49) Which of the following statement is incorrect with reference to frog ?

(1) They eat insects and protect the crop.
(2) Frogs maintain ecological balance.
(3) In some countries muscular legs of frog are used as food by man.
(4) Fertilisation and development are internal.

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50) In frog, right auricle receives blood from :-

(1) Lungs
(2) Truncus arteriosus
(3) Sinus venosus
(4) Carotid artery

51) What is the function of Tympanum in frog ?

(1) Respiration
(2) Excretion
(3) Circulation
(4) Hearing

52)

Mid brain of frog is characterised by the presence of :

(1) One pair of optic lobe
(2) Two pair of optic lobe
(3) Single optic lobe
(4) Corpora qudrigemina

53) In forelimb and hindlimbs of frogs how many digits are present respectively ?

(1) Five and four
(2) Four and five
(3) Five and five
(4) Four and four

54) Alimentary canal in frog terminates into :

(1) Cloaca
(2) Stomach
(3) Ileum
(4) Duodenum

55) In frog Bidder's canal is present in :

(1) Testis
(2) Cloaca
(3) Kidney
(4) Urinary bladder

56) Select the correct statements with regards to excretion of cockroach and choose the correct option which includes all the correct ones :
I. Excretion is performed by malpighian tubules.

II. Cockroach is ureotelic animal.
III. Malpighian tubules are located at the junction of mid gut and hind gut.
IV. Malpighian tubules are lined by glandular and ciliated epithelium.

(1) I, II and III
(2) I, III and IV
(3) II, III and IV
(4) I, II and IV

57) **Statement-I :** The blood vascular system of cockroach is of open type including heart and well developed blood vessels.
Statement-II : In cockroach, blood from sinuses enter heart through ostia and is pumped anteriorly to sinus again.
Choose the correct answer from the options given below :

(1) Both statement-I and statement-II are incorrect.
(2) Statement-I correct but statement-II is incorrect.
(3) Both statement-I and statement-II are correct.
(4) Statement-I is incorrect but statement-II is correct.

58) Match the List-I with List-II related to digestive system of cockroach.

	List-I		**List-II**
(A)	Structures used for grinding of food	(I)	Pharynx
(B)	Located at the junction of foregut and midgut	(II)	Hepatic caeca
(C)	Located at the junction of midgut and hindgut	(III)	Malpighian tubules
(D)	Mouth opens into	(IV)	Mandibles

(1) A-IV, B-II, C-III, D-I
(2) A-I, B-II, C-III, D-IV
(3) A-IV, B-III, C-II, D-I
(4) A-III, B-II, C-IV, D-I

59) **Assertion :** The monitoring of the environment in cockroach is performed by antennae.
Reason : Antennae have sensory receptors.

(1) Both Assertion and Reason are correct & Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are correct but Reason is not correct explanation of Assertion.
(3) Assertion is correct but Reason is incorrect.
(4) Assertion and Reason both are incorrect.

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60) Select the incorrect statement about reproductive system of cockroach -

(1) Female cockroach deposit ootheca in dark and relatively low humidity places.
(2) Development of cockroach is paurometabolous.
(3) Small tubules secrete nutritive fluid in spermatophore.
(4) Ejaculatory duct form 2nd layer of spermatophore.

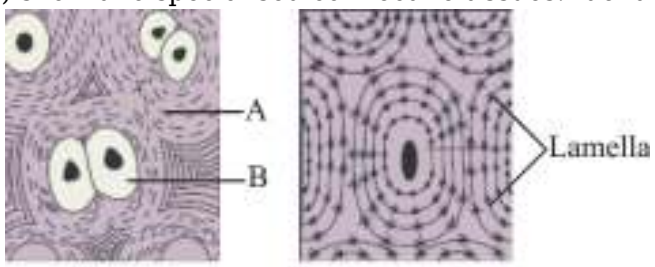
61) In cockroach the first pair of wings arise from ?

(1) Mesothorax
(2) Metathorax
(3) Prothorax
(4) All of these

62) In cockroach exchange of gases take place at the :

(1) Tracheoles by diffusion
(2) Trachea
(3) Tracheoles by active transport
(4) Spiracles

63) Given figures (I and II) show two specialised connective tissues. Identify the figures I, II and the parts labelled as A and B.



	I	II	A	B
(1)	Cartilage	Bone	Collagen fibres	Chondrocyte
(2)	Cartilage	Bone	Collagen fibres	Chondroclast
(3)	Bone	Cartilage	Microtubules	Osteoblast
(4)	Bone	Cartilage	Collagen fibres	Osteoblast

(1) 1
(2) 2
(3) 3
(4) 4

64) **Assertion (A) :** Our heart consist of all 4 type of tissue i.e. epithelial, connective, muscular & neural tissue.

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Reason (R) : Each organ in our body is made up of one or more type of tissue.

(1) A & R both are correct but R is not correct explanation of A.

(2) A is true but R is false.

(3) A is false but R is true.

(4) A & R both are correct and R is a correct explanation of A.

65) Match column-I with column-II correctly and select the correct option given below :

	Column-I		**Column-II**
(a)	Adipose tissue	(i)	Attach one bone to another
(b)	Areolar tissue	(ii)	Specialised to store fats.
(c)	Tendons	(iii)	Attach skeletal muscles to bones
(d)	Ligaments	(iv)	Serves as a support frame work for epithelium.

(1) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

(2) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(3) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

66) Read the following statements carefully and select the correct option given below.
Statement-I :- In unicellular organism, all functions like digestion, respiration and reproduction are performed by a single cell.
Statement-II :- The body of a simple organism like hydra is made of different types of cells and the number of cells in each type can be in thousands.

(1) Statement-I is correct but statement-II is incorrect.

(2) Statement-I is incorrect but statement-II is correct.

(3) Both statement-I and statement-II are correct.

(4) Both statement-I and statement-II are incorrect.

67) Which statement is correct about connective tissue.

(1) Connective tissue is most abundant and widely distributed in simple animals.

(2) Connective tissue include only bones and cartilages.

(3) All connective tissue including blood secrets fibres.

(4) Connective tissue named because of their function of linking and supporting other organ.

68) Tendon and ligament are example of

(1) irregular connective tissue

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(2) regular connective tissue
(3) specialised connective tissue
(4) loose connective tissue

69) Ciliated columnar glandular epithelium is found ?

(1) PCT of nephron
(2) Bronchioles
(3) Airsac
(4) All of the above

70) A pathologist sees cells embedded in a rigid, mineralised matrix. Which tissue is this?

(1) Cartilage
(2) Areolar tissue
(3) Bone
(4) Adipose

71)

Origin of connective tissue :-

(1) Ectodermal
(2) Mesodermal
(3) Endodermal
(4) All of the above

72) **Assertion (A) :** Molluscs exhibit a high degree of body specialisation with distinct head, visceral hump and foot, a soft and spongy layer of skin forms a mantle over the hump.
Reason (R) : The presence of radula in mouth of molluscs allow efficient respiration in adverse environment.

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) **(A)** is true but **(R)** is false.
(3) **(A)** is false but **(R)** is true.
(4) Both **(A)** and **(R)** are true and **(R)** is the not correct explanation of **(A)**.

73) Which of the following option is correct for given diagram regarding their name and character ?

(A)  (B) 

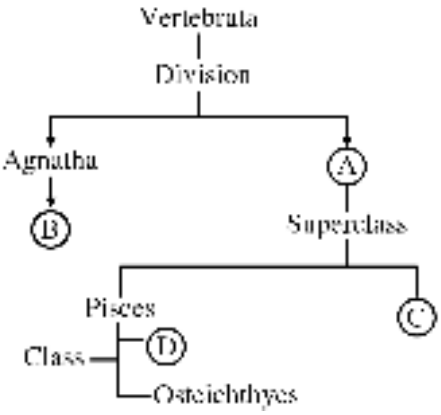
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(1) (A) Hippocampus - Heart 3 chambered.
(B) Chameleon - Fertilization internal.

(2) (A) Hippocampus - Parental care found.
(B) Chelone - Covered by dry cornified skin.

(3) (A) Hippocampus - Placoid scales.
(B) Testudo - Homothermous.

(4) (A) Neophron - fertilization external.
(B) Calotes - tympanum represent ear.



74) Find out the missing lebels in given flow diagram :

(1) (A) - Gnathostomata, (B) - Cyclostomata,
(C) - Tetrapoda, (D) - Chondrichthyes

(2) (A) - Gnathostomata, (B) - Cephalochordata,
(C) - Chondrichthyes, (D) - Osteichthyes

(3) (A) - Urochordata, (B) - Cyclostomata,
(C) - Tetrapoda, (D) - Chondrichthyes

(4) (A) - Gnathostomata, (B) - Tetrapoda,
(C) - Cyclostomata, (D) - Chondrichthyes



75) Select the incorrect option for given figure of animal

(1) Digestion is both extra cellular and intra cellular

(2) Bioluminescence is well-marked

(3) Diploblastic with tissue level of organisation

(4) Fertilisation is internal

76) **Statement - I :** Hagfish which is a jawless fish, lay egg in fresh water and whose ammocoetes larva after metamorphosis return to ocean.
Statement - II : All cyclostomes do not possess Jaw and paired fins.

(1) Both Statement-I and Statement-II are correct.

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(2) Statement-I is correct and Statement-II are incorrect.
(3) Statement-I is incorrect and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

77) How many of the following are included in Echinodermata.
Star fish, Sea-urchin, Sea-cucumber, Sea-hare, Sea-horse, Sea-lily

(1) Two
(2) One
(3) Four
(4) Three

78) Choose the correct pathway of water flow in water transport system :-

(1) Osculum → Spongocoel → Ostia → Exterior
(2) Spongocoel → Ostia → Osculum → Exterior
(3) Ostia → Spongocoel → Osculum → Exterior
(4) Medriporite → Ring canal → Radial canal →Tube feet

79) Which among the following is incorrect about Pisces ?

(1) Their heart is 2 chambered with close single circulation.
(2) They have a streamlined body with paired fins that helps in movement.
(3) Their skeleton is made of cartilage and bone.
(4) They are cold blooded animals and can regulate the body temperature.

80) Identify the incorrect statement :

(1) Ctenophores are marine animals with comb plates.
(2) Platyhelminthes have a flat body and exhibit bilateral symmetry.
(3) Annelids are pseudocoelomates with a segmented body.
(4) Arthropods are characterized by the presence of jointed appendages.

81) Birds and mammals have :-

(1) Three chambered heart with double circulation
(2) Four chambered heart with double circulation
(3) Two chambered heart with single circulation
(4) Four chambered heart with truncus arteriosus

82) Animal that usually undergoes external fertilisation with indirect development is–

(1) *Hyla*
(2) *Scoliodon*
(3) *Taenia*
(4) *Sycon*

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83) In which animal, notochord is present only in larva's tail and absent in adult stage ?

(1) *Ascidia*
(2) *Branchiostoma*
(3) *Balanoglossus*
(4) *Petromyzon*

84) Which of the following features is present in phylum Arthropoda ?

(1) Jointed appendages
(2) Asegmentation
(3) Chitinous endoskeleton
(4) Parapodia

85) In which of the following phylums the adults are radially symmetrical but larvae are bilaterally symmetrical ?

(1) Arthropoda
(2) Mollusca
(3) Echinodermata
(4) Hemichordata

86) The excretory organ of balanoglossus is :-

(1) Protonephridia
(2) Proboscis gland
(3) Prostate gland
(4) Pyloric gland

87) Which of the following groups exhibit the organ-system level of organization ?

(1) Platyhelminthes, coelenterates
(2) Annelids, Arthropods
(3) Sponges, molluscs
(4) Echinoderms, sponges

88) Fill in the blanks (A) and (B) :
___A___ is a mesodermally derived rod-like structure formed on the ___B___ during embryonic development.

(1) (A) Vertebral column, (B) Ventral side
(2) (A) Notochord, (B) Ventral side
(3) (A) Notochord, (B) Dorsal side
(4) (A) Vertebral column, (B) Dorsal side

89) In which organisms alimentary canal has separate opening as mouth & anus.

(1) Coelenterates

(2) Platyhelminthes

(3) Annelids

(4) Ctenophores

90) Find the incorrect match regarding chordates

	A		B
(1)	Germinal layers	(P)	Triploblastic
(2)	Level of organisation	(Q)	Organ system level
(3)	Coelom	(R)	Coelomate
(4)	Symmetry	(S)	Radial symmetry

(1) 1

(2) 2

(3) 3

(4) 4

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ANSWER KEYS

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PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	3	3	2	3	4	4	2	2	2	1	4	4	4	3	1	3	4	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	4	3	2	4	2	4	3	4	3	2	1	1	2	4	4	4	3	4
Q.	41	42	43	44	45															
A.	1	1	1	3	4															

CHEMISTRY

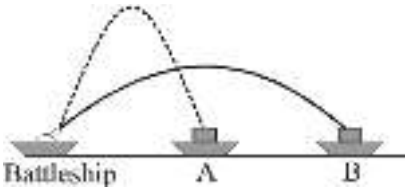
Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	1	2	3	2	4	1	4	1	3	3	4	3	4	1	3	4	2	1	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	1	4	1	3	3	4	3	1	3	3	3	4	3	3	4	2	3	2	3
Q.	86	87	88	89	90															
A.	2	4	2	2	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	2	3	3	2	4	4	3	2	2	4	2	4	3	2	4	2	2	2	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	2	4	2	2	3	3	3	2	4	2	2	2	1	2	1	2	4	2	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	2	4	4	4	1	3	4	3	4	1	2	1	3	2	4	1	1	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	1	1	4	1	3	4	2	2	3	2	2	2	1	4	3	3	3	4	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	1	1	1	3	2	2	3	3	4										

PHYSICS

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1) $H = \frac{u^2 \sin^2\theta}{2g}$

$T = \frac{2u \sin\theta}{g}$

As maximum height for A is more than B, hence its time of flight will also be large compared to time of flight for ship B.

$T_A > T_B$

Hence, ship B will get hit first.

2)

t = 2,4,6 & 8 sec

Slope = 0 ⇒ V = 0

V_{AV} (2s to 6s) = 0 (displacement = 0)

Since slope of the graph is changing continously indicating non-uniform acceleration.

3) ∵ Acceleration is in x-direction only.

∵ Velocity along y-axis will not change.

$= 5^2 = v_y^2 + v_x^2$

$= 5^2 = u_y^2 + v_x^2$

$= 5^2 = 4^2 + v_x^2$

$= v_x = \pm 3 m/s$

∵ $v_x = u_x + a_x t$

$\Rightarrow t = \frac{v_x - u_x}{a_x}$

$= \frac{3-4}{-0.4}$

∵ t = 2.5 sec.

Also, $t = \frac{-3-4}{-0.4}$

$t = \frac{3+4}{-0.4} = 17.5$ sec

4) $a = k\sqrt{t}$

$\int_0^V dV = k \int_{t=0}^{t} \sqrt{t} dt$

$v = \frac{2}{3}kt^{3/2}$

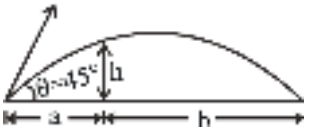
 Average acceleration

$= \frac{\vec{v}_f - \vec{v}_1}{t} = \frac{\frac{2}{3}kt^{3/2}}{t}$

$= \frac{2}{3}kt^{(3/2-1)} = \frac{2}{3}k\sqrt{t} = \frac{2}{3}a$

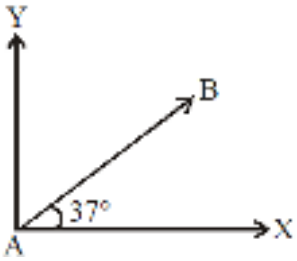
5)

$y = x\tan\left[1 - \frac{x}{R}\right]$



$h = a\tan 45°\left[1 - \frac{a}{a+b}\right]$

$h = \frac{ab}{a+b}$



6)

$\vec{u} = 10\hat{i} + 12\hat{j}$

$u_x = 10$

$u_y = 12$

velocity of wind = $u\hat{i}$

$AB\cos 37° = (u_x + u)t$

$AB\sin 37° = u_y t$

$AB \times \frac{3}{5} = 12t$

$AB \times \frac{4}{5} = (10 + u)t$

$AB = 20t$

$20t \times \frac{4}{5} = 10t + ut$

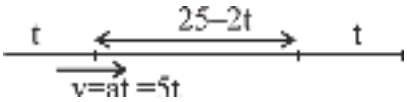
$16t = 10t + ut$

$6t = ut$

$u = 6\ ms^{-1}$

7) Average velocity = $\frac{\text{Total displacement}}{\text{Total time}}$

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or $20 = \dfrac{\frac{1}{2} \times a \times t^2 + (at)(25-2t) + \frac{1}{2} \times a \times t^2}{25}$

$\Rightarrow$ put $a = 5\ m/s^2$

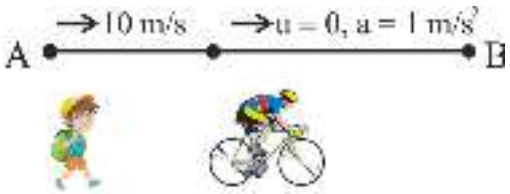
then we get

t = 20 s or t = 5s

t = 20 s not possible

The particle moved uniformly for

(25 – 2t) or 15s.



8)



Let the boy meet the cyclist at B.

Boy (uniform motion) : $10t = 48 + d$

Cyclist (acceelerated motion : $d = \frac{1}{2}.1.t^2$

$10t = 48 + \dfrac{t^2}{2} \Rightarrow t^2 - 20t + 96 = 0$

(t–8) (t–12) = 0

The boy meets the cyclist at t = 8s, 12s

At t = 8s, velocity of cyclist, $v_c = 1 \times 8 = 8 m/s$

$v_b > v_c$, boy crosses the cylist

At = 12s, velocity of cyclist, $v_c = 1 \times 12 = 12$ m/s

$v_c > v_b$, cyclist crosses the boy and will be always ahead

9) $\vec{A} = 10\cos 53^\circ \hat{i} + 10\sin 53^\circ \hat{j}$

$= 6\hat{i} + 8\hat{j}$

$\vec{B} = -10\cos 37^\circ \hat{i} + 10\sin 37^\circ \hat{j}$

$= -8\hat{i} + 6\hat{j}$

$\vec{C} = -10\hat{j}$

$\vec{R} = \vec{A} + \vec{B} + \vec{C} = -2\hat{i} + 4\hat{j}$

10) $\vec{v}_1 = \hat{i} + x\hat{j} + 3\hat{k}$

$\vec{v}_2 = 4\hat{i} + (4x-2)\hat{j} + 2\hat{k}$

$2|\vec{v}_1| = |\vec{v}_2|$

$\left(2\sqrt{1+x^2+9}\right)^2 = \sqrt{16 + (4x-2)^2 + 4}$

$4(10 + x^2) = 20 + (4x - 2)^2$

$40 + 4x^2 = 20 + 16x^2 + 4 - 16x$

$12x^2 - 16x - 16 = 0$

$3x^2 - 4x - 4 = 0$

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$3x^2 - 6x + 2x - 4 = 0$
$3x(x - 2) + 2(x - 2)$
$(3x + 2)(x - 2)$
$3x + 2 = 0$
$3x = -2$
$x = \frac{-2}{3}$
then n = – 2

11) $\vec{a} \cdot \vec{b} = ab\cos\theta = 0$
$\theta = \pi/2$
$\vec{a} \cdot \vec{b} = 4$
$(1)(4)\cos\theta = 4 \Rightarrow \cos\theta = 1, \theta = 0°$
$\vec{a} \cdot \vec{b} = -12$
$(3)(4)\cos\theta = -12$
$\cos\theta = -1 \Rightarrow \theta = \pi$
$\left|\vec{a} + \vec{b}\right| = 5$
$\Rightarrow \sqrt{3^2 + 4^2 + 2(3)(4)\cos\theta} = 5$
$\cos\theta = 0°$
$\theta = \pi/2$

12)

$$\frac{1}{\pi}\int_0^{2\pi} (\sin^2 x)dx$$

$\cos 2x = 1 - 2\sin^2 x$

$$\sin^2 x = \frac{1-\cos 2x}{2}$$

$$\frac{1}{\pi}\int_0^{2\pi}\left[\frac{1-\cos 2x}{2}\right]dx$$

$$\frac{1}{2\pi}\int_0^{2\pi} dx - \frac{1}{2\pi}\int_0^{2\pi}\cos 2x.dx$$

$$\left[\frac{1}{2\pi}x - \frac{1}{2\pi}\frac{\sin 2x}{2}\right]_0^{2\pi}$$

$$\frac{1}{2\pi}\left[x - \frac{\sin 2x}{2}\right]_0^{2\pi}$$

$$\frac{1}{2\pi}[2\pi - 0] = 1$$

13)

$$\text{Area} = \int_0^3 x^2 dx = \left[\frac{x^3}{3}\right]_0^3 = 9$$

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14) $[m] = [v^a g^b \rho^c]$
$M^1L^0T^0 = (LT^{-1})^a(LT^{-2})^b(ML^{-3})^c$
$\Rightarrow a + b - 3c = 0$ .....(1) comparing L
$a + 2b = 0$ .....(2) comparing T
$C = 1$ .....(3) Comparing M
From (1), (2) and (3)
$a = 6, b = -3, c = 1$
$a + b + c = 4$

15)

$[E] = mL^2T^{-2}$

$$K_2 = K_1\left[\frac{m_1}{m_2}\right]^a\left[\frac{L_1}{L_2}\right]^b\left[\frac{T_1}{T_2}\right]^c$$

$$= 10\left[\frac{1kg}{100gm}\right]^1\left[\frac{1m}{200cm}\right]^2\left[\frac{1\,sec}{5\,sec}\right]^{-2}$$

$$= 10\left[\frac{1000gm}{100gm}\right]\left[\frac{100\,cm}{200\,cm}\right]^2[5]^2$$

$$= 10 \times 10 \times \frac{1}{4} \times 25 = 625$$

16) $P = k\rho^a V^b d^c \mu \Rightarrow [P] = [\rho]^a [\,] [V]^b [d]^c [\mu]$
$\Rightarrow ML^{-1}T^{-2} = (ML^{-3})^a(L)(LT^{-1})^b(L)^c(ML^{-1}T^{-1})$
$a + 1 = 1 \Rightarrow a = 0$
$-b - 1 = -2 \Rightarrow b = 1$
$-3a + 1 + b + c - 1 = -1$
$\Rightarrow b + c = -1 \Rightarrow c = -2$
$a + b - c = 0 + 1 - (-2) = 3$

17) When a quantity is derived from others raised to certain powers, the maximum percentage error in the final quantity is the sum of the percentage errors of the individual quantities, each multiplied by the magnitude of its respective power.

$$\frac{\Delta P}{P} \times 100\% = \left(2\frac{\Delta a}{a} + 3\frac{\Delta b}{b} + \frac{\Delta c}{c} + \frac{1}{2}\frac{\Delta d}{d}\right) \times 100\%$$

$$=2(1\%) + 3(2\%) + 3\% + \frac{1}{2} \times 4\% = 13\%$$

18) $T = 2\pi\sqrt{\frac{\ell}{g}}$

$$g = \frac{4\pi^2\ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T}$$

$$= \frac{0.1}{25} + \frac{2 \times 1}{50}$$

$\frac{\Delta g}{g} = 4.4\%$

19) $E = \frac{hc}{\lambda} \Rightarrow [hc] = [E\lambda] \Rightarrow [hc] = [M^1L^3T^{-2}]$

[F] $= \left[\frac{e^2}{\in_0 r^2}\right] \Rightarrow \left[\frac{e^2}{\in_0}\right] = \left[Fr^2\right] = \left[M^1L^3T^{-2}\right]$

$\Rightarrow \left[\frac{e^2/\in_0}{hc}\right] = \left[\frac{M^1L^3T^{-2}}{M^1L^3T^{-2}}\right]$

= Dimensionless

20) From fig. (a) : 4 MS divisions are equal to 5 Vernier scale divisions

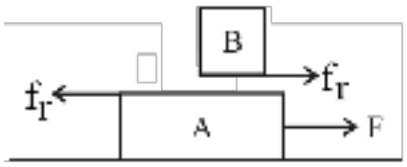
4 MSD = 5VSD; 1 MSD = 0.1 cm

LC $= \left(\frac{5-4}{5}\right) 0.1\,\text{cm} = 0.02\,\text{cm}$

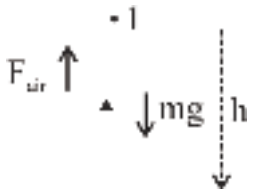
From fig (b) : Measurement ⇒ MSR = 4.1 cm

VSR = 7 × 0.02 cm = 0.14 cm

Reading R = 4.1 cm + 0.14 cm = 4.24 cm

21) 

Angle between f_r and displacement for block B is 0° so work done by f_r is +ve on block B.

22) 

$K_1 = 0$; $K_2 = \frac{1}{2} \cdot \left(\frac{2}{1000}\right) .(30)^2 = 0.9\,\text{J}$

$W_g = mgh = \frac{2}{1000} \times 10 \times (0.5 \times 1000) = 10\,\text{J}$

W_{air} = ?

According to WET,

$W_g + W_{air} = K_2 - K_1$

10 J + W_{air} = 0.9 – 0

W_{air} = –9.1 J

23) $\int dW = -\int F\,dx$

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$$W = -\int_{0.1}^{0.2} (-5x - 16x^3)dx$$

$= 8.1 \times 10^{-2}$ J

24) $KE \propto p^2$

Let $p_i = p$

$$p_f = p + \frac{25P}{100} = \frac{5}{4}p$$

$$\frac{KE_f - KE_i}{KE_i} \times 100\% = \frac{p_f^2 - p_i^2}{p_i^2} \times 100\%$$

$= 56\%$

25)

$P = FV$

$P = ma\nu$ $\nu = u + at = at$

$= (ma)at$

$P = ma^2t$

$P \propto t$

26) $F_s = ma$

$f_L = \mu mg$, $F_s \leq F_L$

$\mu mg = ma_{max}$

$a_{max} = \mu g$

$= 0.15\ (10)$

$= 1.5\ m/s^2$

27) When elevator is moving with uniform speed

$T = F_g$

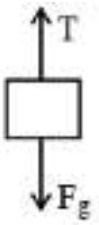
Statement-II

When elevator is going down with increasing speed, its acceleration is downward.

Hence

$$W - N = \frac{W}{g} \times a$$

$$N = W\left(1 - \frac{a}{g}\right)$$ i.e. less than weight



28) $F = mnv = 150 \times 10^{-3} \times 20 \times 800 = 2400$ N.

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29)

average speed = (total distance) / (d_1/v_1 + d_2/v_2)
Take total distance = s unit, then d_1 = 2s/5, d_2 = 3s/5, compute times and average speed.

$$v_{avg} = \frac{s}{\frac{\frac{2}{5}s}{2} + \frac{\frac{3s}{5}}{3}} = \frac{1}{\frac{1}{5} + \frac{1}{5}} = 2.5\,m/s$$

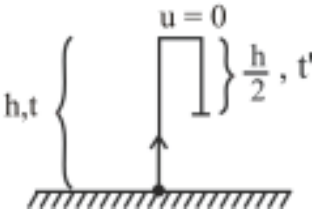
30)

rate of change of speed = component of $\vec{a}$ in direction of $\vec{v}$

$$= \frac{\vec{a}.\vec{v}}{v}\hat{v} = \frac{6+8}{(10)} \times \frac{6\hat{i}+8\hat{j}}{10}$$
$$= \frac{14}{100} \times 2(3\hat{i}+4\hat{j}) = \frac{7}{25}(3\hat{i}+4\hat{j})$$

31)

Time taken to reach to maximum height is $t = \sqrt{\frac{2h}{g}}$
time will be the sum of the time to reach H and the time to fall H/2



Let the time t given in the problem is the time to reach the maximum height H.
(i.e., the time taken while returning)

$$t' = \sqrt{\frac{2}{g}\frac{h}{2}}$$
$$t' = \sqrt{\frac{h}{g}} = \frac{t}{\sqrt{2}}$$

The total time is the sum of the time to reach maximum height (t) and the time to fall to half the maximum height (t'):

Total time $= t + t' = t + \frac{t}{\sqrt{2}}$
$$= t\left(1 + \frac{1}{\sqrt{2}}\right)$$

32) $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$
$\tan\theta = \sqrt{3} \Rightarrow \theta = 60^\circ$
$\frac{g}{2} = \frac{g}{2(u\cos\theta)^2}$
$\Rightarrow u\cos\theta = 1$

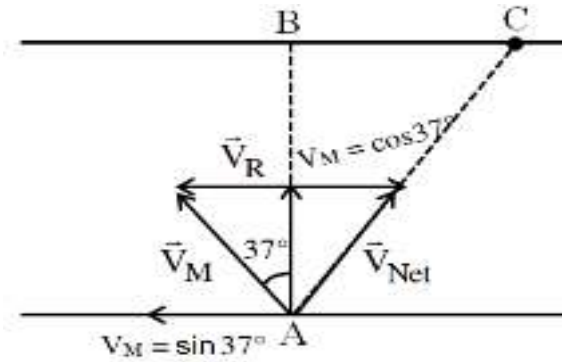
⇒ u cos 60º = 1

33) $\vec{V}_A = 3\hat{i} - 8t\hat{j}$
$\vec{V}_B = 4t\hat{i} + \hat{j}$
at t = 1 sec
$\vec{V}_A = 3\hat{i} - 8\hat{j}$
$\vec{V}_B = 4\hat{i} + \hat{j}$
$\vec{V}_{B,A} = (4\hat{i} + \hat{j}) - (3\hat{i} - 8\hat{j})$
$= \hat{i} + 9\hat{j}$

34)

$(u_{AB})_x = (u_A)_x - (u_B)_x$
$= 10\sqrt{2}\cos 45° - (-10) = 20$ m/s
If 't' is time after which collision occur then :
t × 20 = S
Now, $(u_{AB})_y = 10$ m/s
Also $t = \frac{S_y}{(u_{AB})_y} = \frac{40-20}{10} = 2$ sec.
 S = t × 20 = 2 × 20 = 40 m

35)



As $V_X = V_M \sin 37° = 5 \times \frac{3}{5} = 3$ km/h
which is less river flow (V_R), so man will reach in the downstream point (C)
For crossing time
$t = \frac{AB}{V_M \cos 37°} = \frac{6}{\left(5 \times \frac{4}{5}\right)} = 1.5$ hrs
Now displacement of along the river flow
Shift = $(V_R - V_M \sin 37°)\, t$
Shift = (4 – 3) (1.5) = 1.5 km

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$F_{net} = m \times a$

$F - \mu mg = ma$

$\frac{200}{30} - \mu g = a$

$v = u + at$

$\therefore \quad 4 = 0 + a \times 2$

$\therefore \quad a = 2 \text{ m s}^{-2}$

$\frac{200}{30} - \mu \times 9.8 = 2$

$\frac{140}{30} = 9.8\mu$

$\therefore \mu = \frac{14}{3 \times 10} = 0.47$

36)

37) $T_1 = m_A g$ and $T_2 = \frac{2 \times m \times 3m}{4m} g$ but $T_1 = 2T_2$ so $m_A = 3m$

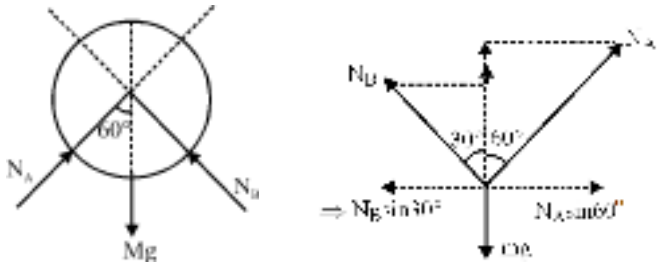
38) $a = \frac{18}{6} = 3 \text{m/s}^2$

2 kg block moves by friction hence

f = ma = 2 × 3 = 6N

39) $a_{\text{after 50 s}} = \frac{F}{M_{\text{after 50 s}}} = \frac{v\frac{dm}{dt}}{1500 - 10 \times 50}$

$= \frac{5000 \times 10}{1000} = 50 \text{m/s}^2$



40)

$N_A \sin 60° = N_B \sin 30°$

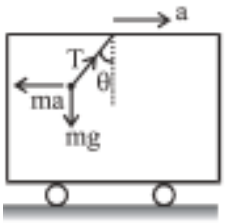
$N_A \cdot \frac{\sqrt{3}}{2} = N_B \frac{1}{2}$

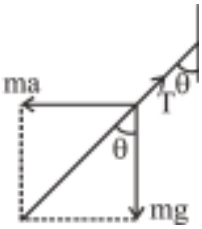
$\sqrt{3} N_A = N_B$

$\boxed{\sqrt{3} N_A = N_B}$

41) The situation is shown in figure.

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When the car moves towards right with acceleration a the rope carrying the mass makes an angle θ.

Acceleration a = $\frac{1}{3}\left(5-\frac{5}{3}\right) = \frac{10}{9}\text{m/s}^2$

$\Rightarrow \tan\theta = \frac{ma}{mg} = \frac{a}{g} \Rightarrow \frac{10}{9\times 10} = \frac{1}{9}$

42)

$P_{out} = \Delta KE/t \cdot P_{in} = 150\,W.$

Calculation:

$W_{out} = \frac{1}{2}m\left(v_f^2 - v_i^2\right) = \frac{1}{2}(10kg)\left(10^2 - 4^2\right)$

= 5(84) = 420 J

P_{out} = 420 J/4s = 105W

$\eta = \frac{105\,W}{150\,W}$ = 0.70 = 70%

43)

Initial KE : $\frac{1}{2}mv_0^2$

Work by gravity = – mgh

work by friction = –μmgd

Final KE : 0

Apply work energy theorem :

$-mgh - \mu mgd = 0 - \frac{1}{2}mv_0^2$

$gh + \mu gd = \frac{1}{2}v_0^2$

Calculation

10(1.1) + 0.6(10) d = 18

11 + 6d = 18

6d = 7 ⇒ d = 7/6

44)

When the ball is released from the top of tower then ratio of distances covered by the ball in first, second and third second

$h_I : h_{II} : h_{III}$ = 1 : 3 : 5 : [Because $h_n \propto (2n - 1)$]

 Ratio of work done $mgh_I : mgh_{II} : mgh_{III}$ = 1:3:5

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45)

$a_r = \frac{u^2}{2s} = \frac{(80)^2}{2 \times 0.5} = (6400)\ m/s^2$

$v^2 = u^2 - 2a_r s'$

$= (80)^2 - 2 \times (6400) \times 0.25$

$v^2 = 3200$

$\boxed{v = 40\sqrt{2}\ m/s}$

CHEMISTRY

46)

16.4215 – 6.01 = 10.415

= 10.41

Digit 1, 0, 4, 1

So total significant figures = 4

47) Mole of $CH_4 = \frac{25}{16}$

No. of CH_4 molecules

$= \frac{25}{16} \times 6.02 \times 10^{23}$

$= 9.41 \times 10^{23}$

1 molecule of CH_4 contains one carbon atom & 4 H-atom

No. of C-atom $= 9.41 \times 10^{23} \times 1 = 9.41 \times 10^{23}$

No. of H-atom $= 9.41 \times 10^{23} \times 4 = 37.64 \times 10^{23}$

48)

	$2NaI$	+	Br_2	→	$2NaBr$	+	I_2
t = 0 mole	0.01		0.01		0		0
after reaction	0				0.01		$\frac{0.01}{2}$

So mole of NaBr produced = 0.01

$W_{NaBr} = 0.01 \times 103 = 1.03$ gm

% yield $= \frac{0.412}{1.03} \times 100 = 40\%$

49)

$CaCO_3 \rightarrow CaO + CO_2$



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$x = \frac{11 \times 100}{44} = 25g$

amount of pure $CaCO_3 = y \times \frac{50}{100} = 25g$

 amount of impure $CaCO_3$ y = 50 g

50) $n_{PbO} = \frac{6.5}{223} = 0.029$

$n_{HCl} = \frac{3.2}{36.5} = 0.087$

PbO + 2HCl → $PbCl_2 + H_2O$

0.029 mol 0.087 mol

(L.R.)

 0.029 mol PbO will form 0.029 mole of $PbCl_2$

51) Freezing point of water is 0°C

$F = \frac{9}{5}(°C) + 32$

 32F

52) (a) (I) 0.5 mole O_3 = 24 g O_3;

(II) 0.5 g atom of oxygen = 8 g

(III) $\frac{3.011 \times 10^{23}}{6.022 \times 10^{23}} \times 32 = 16g\,O_2$

(IV) $\frac{5.6}{22.4} \times 44gCO_2 = 11gCO_2$

53)

	I	O
Moles	$\frac{254}{127}$	$\frac{80}{16}$
	= 2	= 5

I_2O_5

54)

Fe^{+3} is reduced to Fe so it is oxidant and Fe is oxidised into Fe^{+3} so it is reductant.

55)

Reduction : $NO_3^- + 10H^+ + 8e^- \rightarrow NH_4^+ + 3H_2O$

Oxidation Zn → ZN^{2+} + $2e^-$

Multiply oxidation × 4 → $8e^-$ balance.

Combine:

$4Zn + NO_3^- + 10H^+ 4Zn^{2+} + NH_4^+ + 3H_2O$

56) **1. Assertion :** The oxidation state (O.S.) of N in HNO_2 is +3. The O.S. of nitrogen can

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range from –3 to +5. Since +3 is an intermediate state, N can be oxidised (acting as a reducing agent) or reduced (acting as an oxidising agent). The assertion is True.
2. Reason : The O.S. of N varies in different compound (e.g., –3 in NH_3, +5 in HNO_3). The reason is false.

57)

(A) $\overset{+3}{Fe}(\overset{+5}{NO_3})_3 \rightarrow \overset{+2}{Fe}(\overset{+3}{NO_2})_2$
Total e^- gained = 1 + 3 × 2 = 7 (R)
(B) $\overset{+1-2}{Cu_2S} \rightarrow \overset{+2}{Cu^{2+}} + \overset{+4}{S}O_2$
Total e^- lost = 1 × 2 + 6 × 1 = 8 (S)
(C) $\overset{-2}{N_2}H_4 \rightarrow \overset{0}{N_2}$
Total e^- lost = 2 × 2 = 4 (Q)
(D) $\overset{-1}{KI} \rightarrow \overset{+5}{KIO_3}$
Total e^- lost = 6 × 1 = 6 (P)

58)

Option (A) $\underset{+1+5-2}{2KClO_3} \rightarrow \underset{+1-1}{2KCl} + \underset{0}{3O_2}$
atoms of single molecule $KClO_3$ are getting oxidised and reduced, so intra molecular redox.
Option (B) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$.
It is double displacement, so non-redox reaction.
Option (C) $\underset{+2-2}{PbS} \rightarrow \underset{+1-1}{4H_2O_2} + \underset{+2+6-2}{PbSO_4} + \underset{+1-2}{4H_2O}$
Inter molecular redox.
Option (D)
$\overset{0}{P_4} + 3NaOH + 3H_2O \rightarrow \overset{-3}{P}H_3 + 3NaH_2\overset{+5}{P}O_4$
P_4 is getting oxidised and reduced, so disproportionation reaction.

59) **Assertion :-** False, because K_{eq} does not depend on concentration.
Reason :- True
 At equilibrium ⇒ [reactant] ↑ ⇒ reaction shifts in forward direction.

60)

Statement-1: "Solubility of gases in liquid always increases with decrease in pressure" → **False**
(it increases with **increase** in pressure).
Statement-2: "High pressure favors reaction with decrease in gaseous moles" → **True**.

61) (A) $\frac{K_P}{K_C} = (RT)^{\Delta n_g}$
$A_2(g) + 3B_2(g) \rightleftharpoons 2AB_3(g)$; $\Delta n_g = -2$

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$\frac{K_P}{K_C} = (RT)^{-2}$; (A – P)

(B) $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$; $\frac{K_P}{K_C} = (RT)^0$
(B – Q)

(C) $A(s) + 1.5B_2(g) \rightleftharpoons 2AB_3(g)$; $\frac{K_P}{K_C} = (RT)^{-1/2}$
(C – S)

(D) $AB_2(g) \rightleftharpoons AB(g) + 0.5\ B_2(g)$; $\frac{K_P}{K_C} = (RT)^{1/2}$
(D – R)

62) CH_3COOH is weak acid and if we added CH_3COONa then solution is buffer and due to common ion effect acidic character decreases hence pH increases.

63)

HCOOH → It's weak acid
HCOONa → Salt of same acid with strong base

64)

Salt	Classification	Hydrolysis type
NH_4Cl, $(NH_4)_2SO_4$	SAWB	Cationic only
KCl, NaCl, K_2SO_4	SASB	None
NH_4CN	WAWB	Cationic and anionic
$HCOONH_4$	WAWB	Cationic and anionic

65) $S_3 > S_1 > S_2$, in NH_3 odd ion affect, in NaCl common ion affect.

66)

$pK_a + pK_b = 14$ at 25°C only in conjugate pair

67)

$$pH_{neutral} = -\log\sqrt{K_w} = -\log\sqrt{10^{-10}}$$
$$= -\log 10^{-5} = 5$$

Given pH = 5.4 > 5 → solution is basic.

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68) Due to odd ion effect by OH^- in H^+ presence equilibrium shift in right direction so concentration of S^{2-} increases.

69) $M_aX_b \rightarrow aM^+ + bX^-$, $K_{sp} = a^a \times b^b \times s^{(a+b)}$
(a) $Bi_2S_3 \rightarrow 2Bi^{3+} + 3S^{2-} \rightarrow K_{sp} = 108\ s^5 \rightarrow$ (iii) (b) $CdS \rightarrow Cd^{2+} + S^{2-} \rightarrow K_{sp} = s^2 \rightarrow$ (iv)
(c) $Al(OH)_3 \rightarrow Al^{3+} + 3OH^- \rightarrow K_{sp} = 27\ s^4 \rightarrow$ (ii)
(d) $CaF_2 \rightarrow Ca^{2+} + 2F^- \rightarrow K_{sp} = 4s^3 \rightarrow$ (i)

70) 1. NaCl → S.A.S.B → pH = 7
2. NH_4Cl → S.A.W.B → pH < 7
3. Na_2CO_3 → S.B.W.A → pH > 7
4. $CuSO_4$ → S.A. W. B → pH < 7

71) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)\ ;\ \Delta H_1$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)\ ;\ \Delta H_2$

$H_2O(g) \rightarrow H_2O(l)\ ;\ \Delta H = \Delta H_2 - \Delta H_1$

The above process is condensation, so ΔH = – ve ; $\Delta H_2 - \Delta H_1$ = –ve
∵ ΔH_2 and ΔH_1 both are negative,
So $|\Delta H_2| > |\Delta H_1|$

72)

$\Delta H = \Sigma\Delta H_{f(P)} - \Sigma\Delta H_{f(R)}$
or $\Delta H = [81 + 3(-393)] - [9.7 + 3(-110)]$
or $\Delta H = -1098 + 320.3$
or $\Delta H \simeq -778\ kJ\ mol^{-1}$

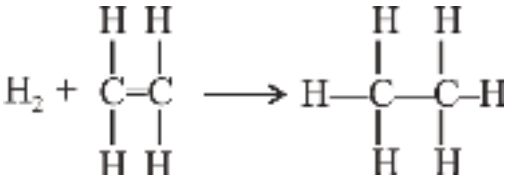
73)

Neutralization of strong acid + strong base = −55.84 kJ/mol.
Neutralization of strong acid + weak base (NH_4OH) = −51.34 kJ/mol.
$\Delta H_{ionisation} = \Delta H_{neut}$ + magnitude of ΔH_{neut} of SASB
= –51.34 + 55.84 = 4.5 kJ

74) $\Delta H_f°\ SO_3 = -98.7 - 298.2$
= –396.9 kJ
$\Delta rH = \Delta H_f\ H_2SO_4 - \Delta H_f SO_3 = -130.2$
$\Delta H_f\ H_2SO_4 = -130.2 - 396.9 - 227.3$
= –754.4 kJ

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75) $\Delta H = 110 + 150 - 80 - 2 \times 100 = -20$ kcal/mol

76)

$\Delta H = \Delta E + \Delta ngRT$
$\Delta H = 17 + 2 \times 2 \times 300 \times 10^{-3}$
$= 17 + 1.2$
$= 18.2$ k cal

77)

(1) $C(s) + O_2(g) \rightarrow CO_2(g)$, $\Delta H_1 = -x$ kJ

(2) $CO(g) + \frac{1}{2}O_2 \rightarrow CO_2(g), \Delta H_2 = -y$ kJ

Target reaction:-

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g), \Delta H_3$

Target Reaction $\rightarrow$ Reaction (1) – Reaction (2)

$\Delta H_3 = \Delta H_1 - \Delta H_2 = -x - (-y)$

$\Delta H_3 = (y - x)$ kJ/mol

78)

$\Delta H = 30$ kJ $mol^{-1} = 30{,}000$ J mol^{-1}
$\Delta S = 75$ J K^{-1} mol^{-1}
$T_{eq} = \Delta H / \Delta S = 30{,}000 / 75 = 400$ K.
For spontaneity, $T > 400$ K.

79) Definition based

80)

At equilibrium $\Delta G = 0$
$\Delta H = -30$kJ $= -30 \times 10^3$ J
$\Delta S = ?$
We know that, for chemical reaction

$$\Delta S = \sum_n nS_{(product)} - \sum_m mS_{(reactant)}$$

$$\Delta S = 50 - -\left(\frac{1}{2} \times 60 + \frac{3}{2} \times 40\right)$$

$$\Delta S = -40 Jk^{-1} mol^{-1}$$

Now, $0 = -30 \times 10^3$ J $- (T \times (-40))$
$0 = -30000$ J $+ 40$ T
$40T = -30000$J

$T = \frac{30000}{40} = 750$

81)

This question is based on $\Delta S \propto \Delta ng$.
The reaction with the largest positive value of Δn_g will have the maximum (most positive) ΔS.
1. Option 1 : $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$
   $\Delta n_g = 2 - (1 + 1) = 0$
2. Option 2 : $N_2(g) + 2O_2(g) \rightarrow N_2O_4(g)$
   $\Delta n_g = 1 - (1 + 2) = -2$
3. Option 3 : $S(s) + \frac{3}{2}O_2(g) \rightarrow SO_3(g)$
   $\Delta n_g = 1 - \frac{3}{2} = -\frac{1}{2}$
4. Option 4 : $2C(s) + O_2(g) \rightarrow 2CO(g)$
   $\Delta n_g = 2 - (0 + 1) = +1$

82)

Intensive properties are not in additive nature.

83)

Question Asking About:
Find the value of ΔE

Given Data:
$\Delta H = -1366.5$ KJ mol^{-1} , $T = 27°C$

Concept:
$T = 27 + 273 = 300$ K , $R = \frac{8.314}{1000}$ KJ mol^{-1} K^{-1}
$\Delta ng = 2 - 3 = -1$
$\Delta H = \Delta E + \Delta ngRT$

Formula:
F

Solution/Explanation/Calculation:
$-1366.5 = \Delta E - \times \frac{8.314}{1000} \times 300$
$-13665 = \Delta E - 2.5$
$\Delta E = -1366.5 + 2.5$
$= -1364$ KJ

84)

$H_2O(\square) \rightleftharpoons H_2O(s)$ is exothermic $\Delta H < 0$

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ΔG = 0 at equilibrium, entropy decreases, volume increases

85)

ΔE = q + w in adiabatic q = 0
So $\boxed{w = \Delta E}$

86)

ΔE = q + w
 Isothermal
ΔE = 0
$\boxed{q = -w}$

87)

In vacuum → w = 0
$\boxed{\Delta E = q + w}$
adiabatic → q = 0
so ΔE = 0

88)

Endothermic processes can be spontaneous

89)

ΔE = q + w
in adiabatic q = 0
so $\boxed{w = \Delta E}$

90)

Assertion - correct
Reason - Incorrect because enthalpy decrease favors spontaneity but is not the only factor.

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91) NCERT Pg # 17-17

92) NCERT Pg.# 14

93) NCERT Pg. No. # 15

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94) NCERT, Pg. # 14, 15

95)

NCERT Pg. # 15, 14, 16, 17

96) NCERT-XI, Pg. # 13

97) NCERT, Pg. # 18

98) NCERT, Pg. # 18

99) NCERT Pg. No. # 18

100) NCERT Pg. # 32

101) NCERT Pg. # 30, 32

102) NCERT Pg. # 26, 27

103) NCERT Pg. No. # 13, 27

104) NCERT-XII Page No. 24

105) NCERT, Pg. # 29

106) NCERT, Pg. # 29

107) NCERT-XI Page No 30, 31

108) NCERT Pg. # 33

109) NCERT Pg. # 4

110) NCERT Pg. # 8

111) NCERT-XI, Pg. # 13

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114) NCERT XI Pg. # 4

115) NCERT Pg # 17, 18

116) NCERT-XI, Pg. # 16

117) NCERT Pg. # 21

118) NCERT Pg. # 18

119) NCERT Page No. 11

120) NCERT Pg. No. # 18

121) NCERT-XI, Pg. # 11

122) NCERT-XI, Pg.# 90

123) NCERT-XII Page No. 32-33

124) NCERT-XI, Pg. # 28

125)

NCERT-XI, Pg. # 25

126) NCERT-XI, Pg. # 32

127) NCERT XI Pg.#29

128) NCERT-XI, Pg. # 29,30

129) NCERT XI Pg.#17,20,21

130) NCERT XI 32

131) NCERT No. Pg.#29

132) NCERT 11_Pg. No. 24,26

133) NCERT-XI, Pg. # 32,33

134) NCERT-XI, Pg. # 14,15

135) NCERT Pg. # 8 (E)/(H)

136) NCERT Pg # 83

137) NCERT Pg. # 121

138)

NCERT-XI, Pg.# 80-81

139) NCERT Pg.# 84

140) NCERT Pg # 118

141) NCERT - 2023 -24 . Pg. 83

142)

NCERT Pg.#119

143)

NCERT-XI, Pg. # 116

144)

NCERT-XI, Pg. # 81

145)

NCERT - 2023 -24 . Pg. 83

146)

NCERT (Old)-XI, Pg. # 114

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148) NCERT Old Pg. No. : 113

149) NCERT Pg.# Module.

150) NCERT Pg.#151,152 (Module).

151) NCERT Pg.# Module.

152) NCERT No. 114

153)

NCERT XI (Old), Pg.# 104

154) NCERT Pg# 80

155) NCERT Pg.#103.

156) New NCERT Pg. # 79

157) NCERT Pg.#102.

158) Module - 77

159) NCERT XI Pg.# 101

160)

NCERT-XI, Pg. # 105

161) Old NCERT
Module

162) NCERT Pg. # 44

163) NCERT Pg. # 48-49

164) NCERT, Pg. # 47

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165) [RATIONALISED NCERT BOOK 2024-2025]
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166) NCERT Pg. # 47

167)

NCERT Pg. No. 45

168) NCERT (XIth) Pg. # 49

169) NCERT Pg. # 47-48

170)

NCERT-XI, Pg.# 43

171)

NCERT-XI, Pg. # 49-50

172)

NCERT-XI, Pg.# 48

173) NCERT Pg. # 55

174) NCERT Pg. # 44

175) NCERT Pg # 45

176) NCERT Pg # 45

177) NCERT_(XI), Pg. # 37

178) NCERT Pg. 39

179) NCERT Pg # 47

180) NCERT, Pg, # 45-46

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